The RISC V Revolution

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Introduction

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4. open-source ISA from scratch

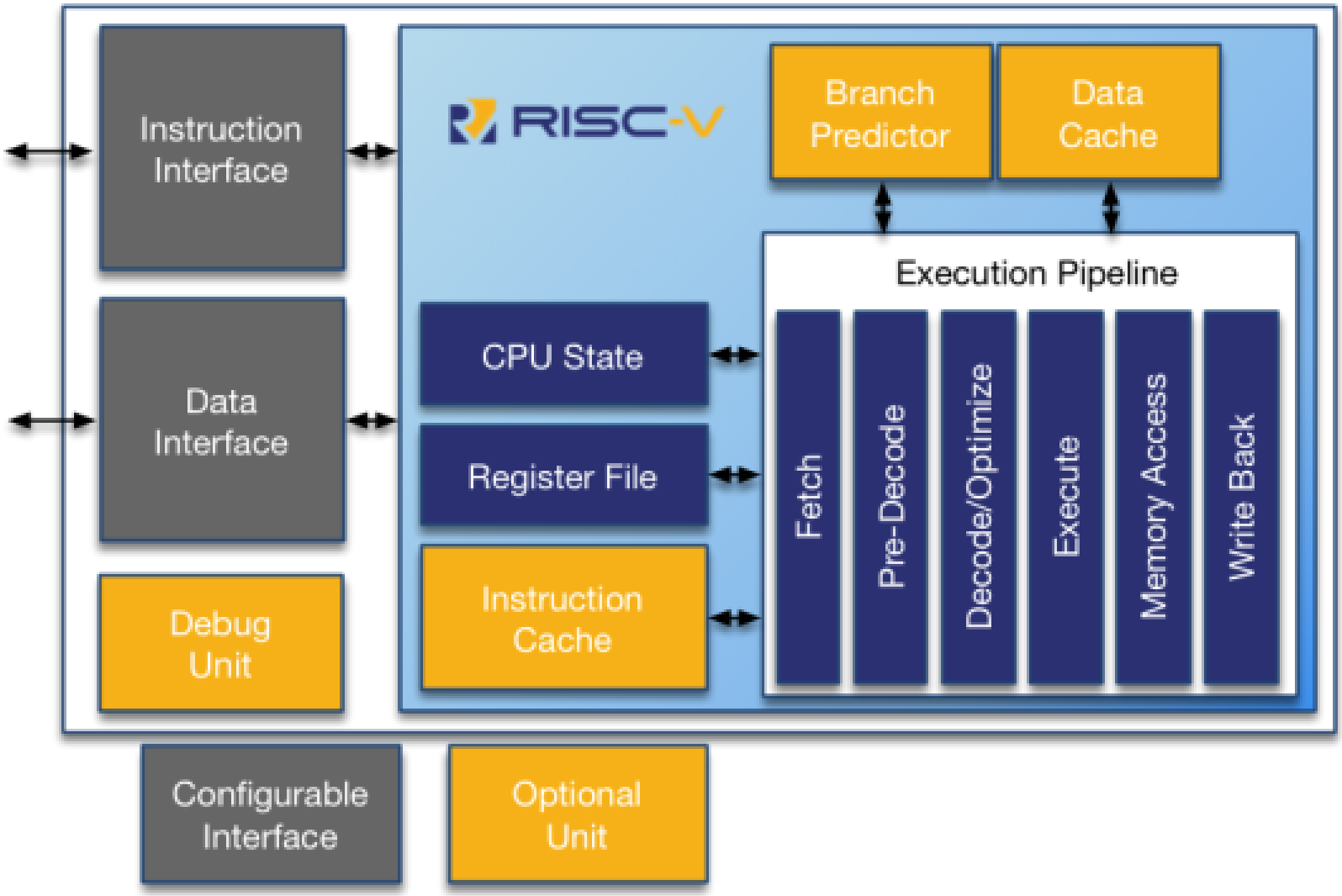
Why RISC V Matters

1. **No Licensing Fees:** Companies and researchers can design custom processor cores without paying royalties to vendor companies.
2. **Geopolitical Neutrality:** Being open-source, it reduces dependence on single corporate vendors or foreign export-control regulations.
3. **Academic & Commercial Adoption:** It is widely used in microcontrollers, hard drive controllers (e.g., Western Digital), AI accelerators, automotive chips, and embedded systems, with growing support for Linux and Android desktop/server platforms.

History of RISC V

1. RISC-V was born in May 2010 at the University of California, Berkeley. Lead researchers Krste Asanović, Andrew Waterman, Yunsup Lee, and computer architecture pioneer David Patterson needed an Instruction Set Architecture (ISA) for research and teaching.
2. ”V” stands for the fifth major RISC architecture designed at UC Berkeley (earlier versions included RISC-I and RISC-II in the 1980s, which helped lay the groundwork for modern processors).
3. RISC-V Foundation was established to manage and standardize the architecture collaboratively,(2015)
4. Transition to RISC-V International (2019–Present)(Switzerland)
5. Adoption by Academia and Industry

Architecture and Features



Modularity

1. The Core Vocabulary (The Base)
2. Extension
   - The Math Helper: Add an extension for fast multiplication and division.
   - The Decimal Helper: Add an extension for handling precise decimal numbers.
   - The AI Accelerator: Add vector extensions to crunch massive amounts of data simultaneously for artificial intelligence.

Hardware Implementations

Category	Key Processors & SoCs	Target Applications	OS Support
Micro controllers	ESP32-C3/C6, WCH CH32V003, SiFive E20	Smart home IoT, sensors, wearables, automation	Bare-metal, FreeRTOS, Zephyr
Edge AI & Vision	Canaan Kendryte K230, Sipeed Maix Series	Smart cameras, robotics, computer vision	Lightweight Linux, RTOS
Application CPUs	StarFive JH7110, SpacemiT K1, SiFive P550	Developer SBCs, laptops, workstations	Debian, Ubuntu, Fedora, Android
Cloud & Datacenter	Sophgo SG2042, Ventana Veyron, Tenstorrent	Hyperscale cloud, HPC, LLM inference	Enterprise Linux (Ubuntu, Fedora)

Indian scenario

- SHAKTI (2018, IIT Madras): Designed for strategic, space, and defense applications;
- AJIT (2018, IIT Bombay): A microprocessor for industrial and robotics applications;
- VIKRAM (2025, ISRO-SCL): A processor developed for space applications such as navigation, guidance, and mission operations; engineered to withstand extreme space conditions;
- THEJAS64 (2025, C-DAC): Designed for industrial automation.
- Dhruva64 (2025 C-DAC)

How to get Started

3 Paths

1. Visual Simulation (Ripes)
2. Embedded Tinkering (ESP32 C3/Raspberry pi pico V2 or SBC Milk V Mars)
3. HDL CPU design (verilog/System verilog)

Conclusion

- **Microcontrollers & Storage**-Complete Takeover-"Billions of chips inside hard drives (Western Digital), smart devices (Espressif), and microcontrollers already run on RISC-V."
- **Edge AI & Automotive**-In Progress-"Widely used for custom AI accelerators (Meta, Tenstorrent) and automotive controllers."
- **Smartphones & PC Desktops**-Early Stages-"Google has made RISC-V a top-tier architecture for Android, but mainstream consumer devices are still a few years away from matching ARM/x86 software ecosystems."
- **Data Centers & Cloud** -Emerging-"Tech giants use RISC-V for specialized cloud hardware and offload processors, though x86/ARM still dominate primary server CPUs."